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Dealing with waste along canals.

Human Centred Design Engineering

Project brief and desk research:

Brief:

Dealing with litter and waste along rivers and canals in London.

***What**: Reduce the amount of litter found along London canals.*

***Who**: Canal & River Trust and LAGER Can.*

***Where**: London.*

***Why**: Not only does litter render areas unsightly and dirty, it is also a major environmental hazard both through the pollution it causes and the harmful chemicals that can be found in many everyday products, such as batteries and cigarettes.*

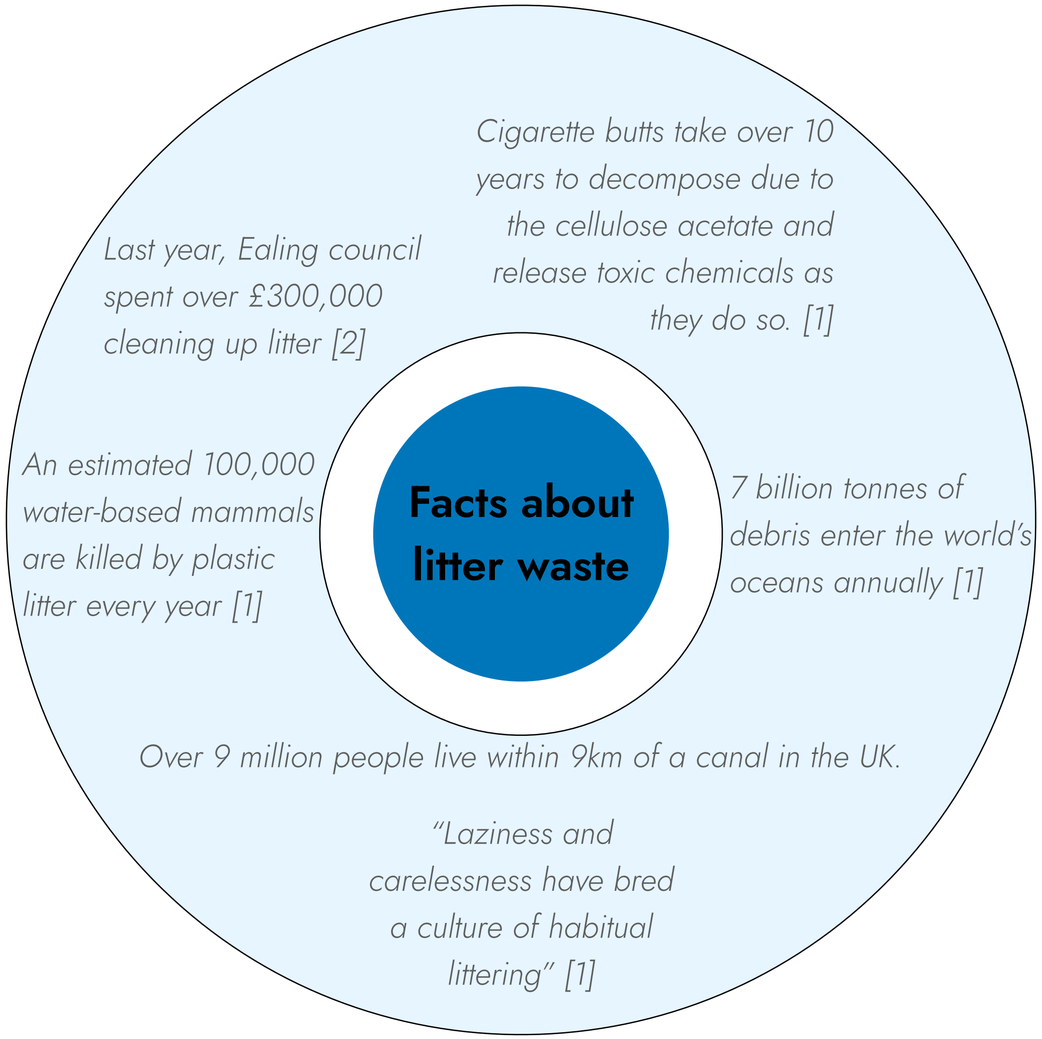
***How**: Begin with desk research and then move onto User interviews and research in order to gain a better understanding of problems they face and ways they might be tackled.*

Canal & River Trust:

The Canal & River Trust looks after a 2000 mile-long, 200 year old network of canals, rivers, reservoirs and docks around the UK [4] which has been operating since 2012. They host many volunteering events to pick up litter in different areas every month as they feel that canals offer both physical and mental health benefits and want to continue protecting them as they are apart of their heritage.

LAGER Can:

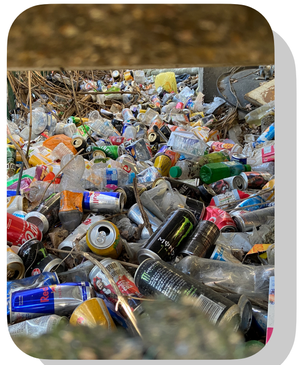
LAGER Can (Litter Action Group for Ealing Residents) is another organisation that has over 2000 members [3] and work on both public spaces like parks and also canals around their borough. While they hold events as well, they also incentivise individual litter picking and deal with fly-tippers: "We're keen to change the behaviours that lead to littering and fly-tipping. We regularly report fly-tippers to Ealing Council. Many have paid large fines." [4]



User research – Overview of the approach:

Observations

To begin, we needed to go out and look at how bad canal waste could get. To do this, we travelled along the canal in Acton and other areas with litter in order to make **assumptions** *about where litter forms, what type of litter forms, and how bad the problem can really get.*



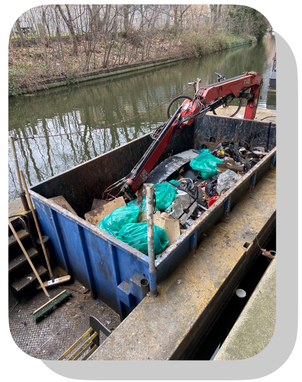
These walks not only led us in working on canals and understanding the problem, but also on us meeting a volunteer for the Canal And River Trust called Crispin.

Interviews with Crispin

Crispin is a volunteer for the Canal and River Trust we met on one of our walks while he was picking up rubbish along the canal. He helped us massively and gave us many **insights** *into general canal waste and how it is dealt with.*

Talking to him, we got many ideas on what we wanted to do next. However, the most helpful thing he gave us was access to the volunteering events for the Canal and River Trust.

Volunteering with the trusts



Taking part in their event where we picked up rubbish was probably the most helpful thing we did. In it, we helped them shovel and transport waste using wheelbarrows to a boat which would then go on to transport it to waste depot. The pile-up had been left there from a homeless person that had lived there for around a year. Therefore, going there helped us in understanding the many problems they have, ranging from problems with volunteering, to the way the waste is picked up and dealt with and even problems in simply picking up the rubbish.

User research – Observations and Crispin:

The observations

Going around the canals, we found that most litter formed in areas around bushes which covered the view from direct sight, areas near exits to the canal, and areas away from bins. This allowed us to make the assumption that: **Litter seems to form in secluded and covered areas.**

In terms of waste that we found, the vast majority of it was **litter** *such as cans, bottles and packaging. However, there were also a few* **cigarettes** *around and every once in a while we would find something larger like a pile of clothes or a lime bike which would definitely be harder to clean up.*

Crispin

The interview with Crispin was extremely helpful due to the large amount of data and information that he gave us:

Types of waste:

Casual litter: (Cans, Paper, Bottles)

Dumped litter: (Clothes, mattresses, kettles)

Stolen property: (Lime bikes, wallets)

Types of litterers:

- *40% try to recycle and almost never litter*
- *40% are a little bit careless and sometimes litter*
- *10% are very careless and litter a lot*
- *10% do not care and litter all the time*

Why do people litter:

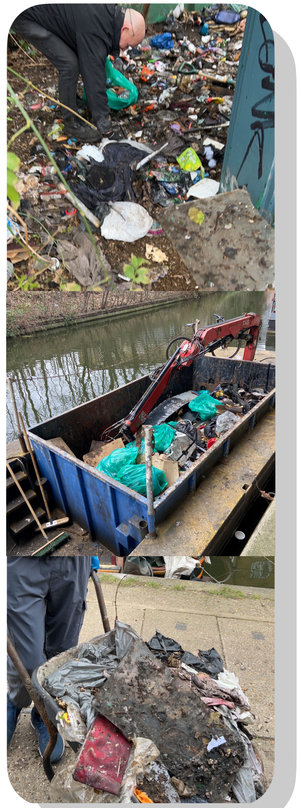
- *The area is* **hidden** *from direct view (such as behind bushes, underneath stairs and in alleys)*
- *A* **lack of bins** *often results in people not wanting to carry trash (Crispin only picks up litters when he goes on walks if he knows there's a bin he can drop the trash off at on the way)*
- *If people* **see litter**, *they are more likely to litter themselves*
- *Fly litterers do so to* **avoid paying** *money to get rid of their waste or to* **avoid travelling** *far*

User research – Canal & River Trust event:

We attended an event near Paddington where we helped clean up an exceptionally bad area while conducting observations and interviewing the other volunteers.

Mark:

*At this event, we met Mark who also works for LAGER Can who happily complained to us about issues he has, such as the company Kingdom Security that is supposed to stop littering in Ealing but does "f*ck-all".*



Greg:

Greg talked to us about the process and why they do things. For example, they have found that if they make the area look clean and looked after then people are much less likely to litter. He wishes that there was a system in place like Singapore that would punish litterers more but knows that's unlikely to occur.

Their process:

1. *Firstly, they identify areas with a large build-up of litter. This is either done by public reports that are emailed to them or volunteers which spot and report the severity and position.*
2. *The next step is to move the welled narrowboat as close to its location as possible in order to facilitate it being filled up and to transport the equipment used for the clean-up including shovels, gloves, wheelbarrows, rakes, etc.*
3. *The waste is then shovelled, picked up, put into wheelbarrows and then unloaded into the well of the narrowboat.*
4. *When the narrowboat is either too full to continue, or the waste is fully cleared, it will then transport the waste to a waste depot that sorts and recycles it if able.*
5. *If there is a need, the vegetation would also be trimmed down or even fully cut down depending on the density of its foliage and how impactful it is on the walkway.*

The problems

Often large build-ups of trash can go **unnoticed** *for a long time due to a lack of reports (the area we cleaned up had been there for at least a year).*

The **speed limit** *on the boats is 4mph (It took them 7 hours to bring the boat to Paddington).*

Little bits of plastic and trash often get **stuck** *and caked in the mud making them rly hard to completely clean up the area.*

Depending on the waste that is being cleaned up, the boat **fills up** *really quickly.*

A **lack of boats** *means they can't work as many areas as they'd like while a* **lack of waste-depots** *means more fly tipping occurs. However, they stressed that the main issue they had was with* **funding and volunteers**.

Insights and opportunities:

Insights:

1. *From the interview with Crispin, we found that he does not pick up rubbish in areas without bins as he does not want to carry the trash all the way home.*
2. *From helping with the volunteering, we found out how hard and labor intensive the process is, but also found a major issue in the mud packing of the trash and it getting stuck.*
3. *From the interviews at the event, we found that they had many issues regarding time management and the lack of areas they could clean for every event.*
4. *The interviews also helped us in understanding that some people litter just because they don't have a better option, often due to a lack of waste depots.*

Opportunities:

1. *If we increased the availability of bins, people might be more likely to pick up and bin the trash or even not litter at all. This means there is an opportunity around a product that either helps people find bins or deal with the litter.*
2. *We could devise a product that helps pick up the pieces of waste that get stuck in the mud and are very hard to remove.*
3. *It may be worth looking into a way to improve their process and make it more time effective so that more areas can be cleaned.*
4. *We could also look into reducing the amount of people who litter and ways to discourage this or even encourage the opposite, to pick up the litter.*

How might we...

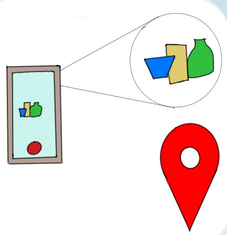
1. *HMW incentivise people to pick up litter?*
2. *HMW effectively pick up small bits of trash that are stuck in the mud?*
3. *HMW make the litter picking process more efficient?*
4. *HMW localise and organise the pick up of trash more efficiently?*
5. *HMW incentivise people to stop / reduce their litter throwing along canals?*

HMW decrease the time taken for the Canal & River Trust's cleaning process around London?

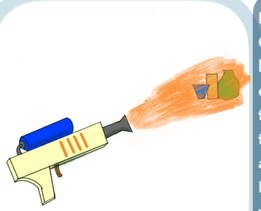
Ideation – 15 ideas:

This page shows the 15 ideas with a brief explanation on the right and the concept score below it. (How they were assessed can be found on the supplementary page)

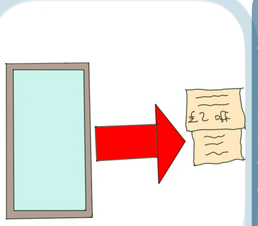
Plough - Helps to break the ground up making it easier to shovel the little bits of dirt up.



App to report litter - Tells the trust areas where there are large build ups of litter so they know where they need to go.



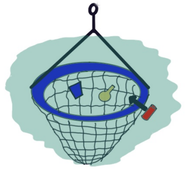
Flamethrower - Obliterates the litter meaning it does not need to be transported anywhere. Helps with vegetation as well.



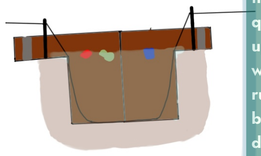
App that incentivises the user to pick up litter - this can be done with vouchers or other similar rewards.

Putting a litter picker next to bins- this might incentivise people to pick up litter on their walks out as they don't need to bring their own equipment.

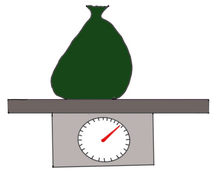
Shovel with rake at front - Helps to break the ground up while shovelling the mud and dirt.



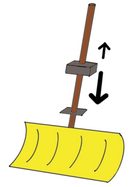
Floating bin which traps floating litter - Has a hook on top to easily pull it up and empty the litter and just floats around.



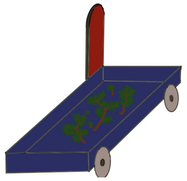
Nets at locks - these can be quickly pulled up together with all the rubbish that builds up and deposited into the boat.



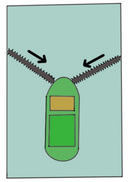
Incentives at landfill sites that reward volunteers for the amount of litter they bring - this can be done through vouchers or other rewards.



Shovel with a sliding weight - can be dropped onto the metal plate driving the shovel into the ground making it easier to dig.

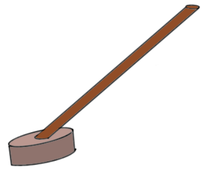


Vegetation cutting robot - has a chainsaw on the side that cuts all vegetation along the path.

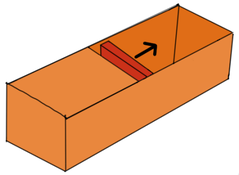


Boat with roller arms - the arms spin pushing all the trash along the canal into the centre of the boat where it can easily be picked up.

Putting trackers on litter-able waste - this means it can be found quickly and it'll be easier to see where stuff has been littered.



Stick on a magnet - picks up metal bits stopping you from getting cut.



Portable trash compactor - helps fit more litter on the boat.

Development of the app to report litter:

Initial idea:

The app should be used to report litter to the trust and so should have a geo-location associated to it. It should also be able to tell the trust how severe the problem is which can be done through image or a form.

What does the trust need to know?

- *How bad the problem is.*
- *The location.*
- *If the vegetation needs to be trimmed back.*
- *If heavy machinery will be required.*
- *If the boat needs to be brought there or the litter can just be picked up using a plastic bag.*
- *If there are any other reported areas nearby they can clean up as well.*
- *If the litter has been reported already.*

Co-design session with Crispin

We ran a Co-design and feedback session with Crispin where we got feedback off him based on our ideas and in which I got him to rank the different possibilities for the app in order of what is needed from most to least so that I could assess what needed to be put on the app. In it, he also added some thing from the list on the left, as can be seen in the list below:

1. **Location of the litter.**
2. **Severity of the waste.**
3. **Are there any toxic materials such as batteries or chemical bottles that can be seen? (Crispin added this)**
4. **Does the vegetation need to be cut back?**
5. **Is heavy machinery required?**

Lo-fi prototyping

To prototype the app, I decided to just create a simple model on some paper and then have a few people test it by running through the motion of using it in order to see what problems they might have:

Feedback from the above:

From the Co-design, I found I would need to include in the form wether the user saw any toxic materials as those would need to be cleaned up quicker and that it was very important for them to see how the severe the problem was so they could assess it.
From the lo-fi prototyping, I found that many users struggle to assess how bad the severity of the waste was without some examples, so it will be helpful to include an extra information page.

Development of the litter picker:

Initial idea:

The litter picker should be available to all the public to use for free so that they can pick up litter while they are out walking without having to use their hands to touch it directly.

What does the product need?

- *The product should be able to pick up litter easily*
- *The product should be attached to the side of bins*
- *The product should be safe for a user to handle*
- *The product should encourage the user to interact with it in order to get more people to pick up litter.*
- *The product should be hard to break as it will be available to the public and so might end up being abused slightly.*

User Feedback

To get some feedback on the idea, I asked Crispin and some other associates on the design. They helped me to find some flaws in the design and so ways in which could be improved, however a glaring issue that someone told me was:

"If I saw that I would try to steal it for sure"

Another issue was that some people might not want to use it as:

"I feel like the handle would be dirty so I don't really want to make my fingers icky"

"I don't think this would really make me want to pick up any litter"

Lo-fi prototyping

To prototype this product, I decided to make a model bin and then attached a O-shaped holder for the grabber on its side. Then I made a simple grabber model out of some pieces of wood and hot glue in order to represent the litter picker.

Feedback from the above:

From the user-feedback on the design, some problems were brought up. As the main one was on stealing, the best solution I could come up with was to put a battery powered tracker in the litter picker, this would mean we would also need some way to charge it which would increase the complexity.

The lo-fi prototyping also brought some problems up. One was that there needs to be some sort of hole in the side of the holder so that the litter picker can be inserted inside, however, that would mean that the litter is likely to fall out, so it would thus be needed to be held in place, possibly using magnets.

Development of the floating bin:

Initial idea:

The floating bin is a way of collecting rubbish along the canal passively just by utilising the flow of the water and having the trash get trapped in its centre.

What does the product need?

- *The bin should be able to float just enough that the floating litter can flow over its top into it.*
- *The bin should not damage any boats that may hit it and should also not be damaged by them.*
- *The bin should trap the waste in its centre.*
- *The bin should have some way to lift it out of the water so that the waste can just be dropped into the well and does not have to be moved piece by piece.*

User Feedback

To get some feedback on the idea, I asked Crispin and some other associates on the design. They helped me to find some flaws in the design and so ways in which could be improved. A problem Crispin was worried about was:

"Could this damage the environment? Such as by rubbing against vegetation or if fish fall into its centre and are not able to get out?"

Another problem which would have to be dealt with was:

"If the bin gets filled with litter, won't it just sink?"

"I don't know if the litter would even go in or it would just push around the edge"

Lo-fi prototyping

Although I could prototype the whole product, what I really needed to work out was the shape the ring on the outside would need to be in order to make litter fall in without it coming back out. To do that, I made a few models of different shapes and gently pushed materials over them in order to see what worked the best:

Feedback from the above:

From the user-feedback on the design, the first issue is something that would require research around a proper model of the product which cannot be conducted now. However, the second one should not even be an issue as the litter is all floating anyways and so won't weight on the bin.
From the lo-fi prototyping, I found the 3rd method to be the best, which means that that will be the one that I will bring forwards.

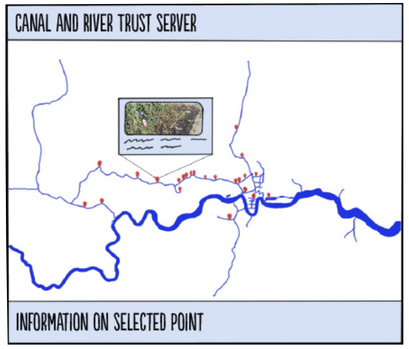
Final concepts:

App to report litter:

1. User takes a geo-tagged picture of the litter or trash.

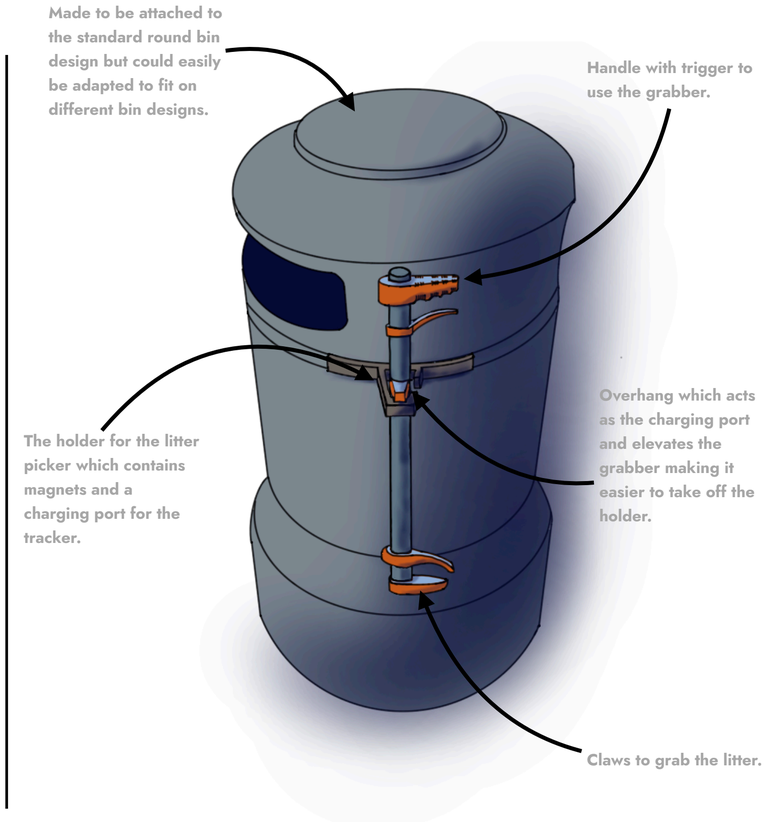
2. They then fill in a quick survey to give some extra information to the trust.

2.1 If the user needs help for the form they press the info button and are brought to this page.

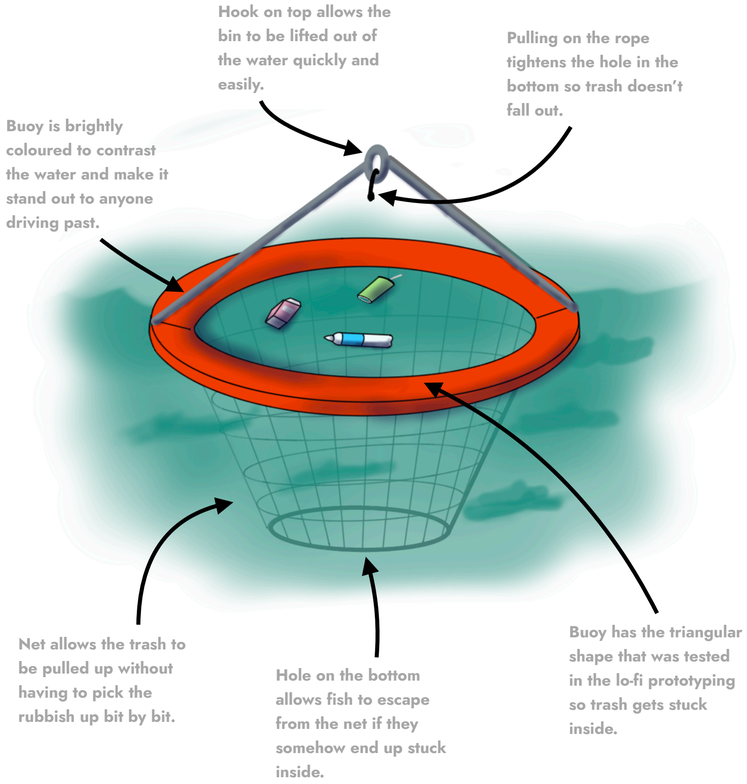


3. The information is then sent to the Canal and River Trust server where they can view each individual point that is sent in on a map together with the data from the survey

Public litter grabber:



Floating bin:



Supplementary page:

Appendix:

[1]: environment.cenn.org
[2]: aroundealing.com
[3]: lagercan.com
[4]: canalrivertrust.org.uk

Concept selection methods:

1. Removing undesirables:

The first step was to remove the ideas which would either not be feasible, already existed, or were just plainly stupid. These were:

- *The plough:*
  - *Has already existed for millennia and so won't be very innovative.*
- *The flamethrower:*
  - *Would be quite dangerous and would probably do more bad than good.*
- *The vegetation cutting robot:*
  - *Would be dangerous as well and probably wouldn't work very well.*
- *The tracker on the litter-able waste:*
  - *Would have to be a government enforced practice.*

2. Scoring ideas:

The ideas were then given scores in different categories in order to bring them down to 5 which the client could then give feedback on. These were:

Product	Desirability	Feasibility	Viability	Effectiveness in context	Total /20
2. App to report litter	5	4	4	4	17
4. Reward app	3	2	2	2	9
5. Litter picker	4	5	4	2	15
6. Shovel with rake	3	5	3	3	14
7. Floating bin	4	4	4	4	16
8. Nets at locks	4	3	4	3	14
9. Landfill rewards	3	2	2	1	8
10. Shovel with sliding weight	3	5	3	2	13
11. Vegetation cutting robot	2	1	1	2	6
12. Boat with roller arms	2	1	1	2	6
14. Stick on a magnet	3	5	3	2	13
15. Portable trash compactor	4	3	3	4	14

3. Client feedback:

Out of the table, I chose the top 5 scoring products and brought them to my client in order to assess what they found to be more helpful.

- *The* **app to report litter** *had very good feedback and so will be brought forwards.*
- *The* **litter picker** *was a bit skeptical as there were worries on people stealing it but it would still be good to develop it.*
- *The nets at locks would unfortunately not work as they would stop the gates from opening and closing.*
- *The* **floating bin** *also had good feedback.*
- *The portable trash compactor seemed good but there were worries about safety and potential legal repercussions in-case someone got injured*